Hyperspectral Anomaly Detection via Locally Enhanced Low-Rank Prior

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***Abstract*— Anomaly detection is an active area of research in hyperspectral information processing. Recently, low-rank representation has been applied in hyperspectral anomaly detection. However, the existing low-rank-based methods either involve a complicated dictionary construction process or the anomaly–background separation which is not sufficient. In this article, to solve these problems, a novel hyperspectral anomaly detection method based on a locally enhanced low-rank prior (LELRP-AD) is proposed. This article is inspired by the observation that, in local homogeneous regions, the background signals hold an enhanced low-rank property while the anomalies exhibit spatial sparsity. Based on this observation, the background pixels can be low-rank reconstructed by a set of basis background signals, whereas anomalies can be represented as sparse residuals. First, image segmentation is performed to enhance the homogeneity of the background, in which a Potts-based image segmentation algorithm is adopted with postprocessing, thus avoiding the need for a complicated spectral dictionary for the representation of the background. Furthermore, the original hyperspectral data matrix is augmented with extracted background endmembers for the low-rank and sparse matrix decomposition, to further achieve anomaly–background separation. The experimental results obtained on four real hyperspectral data sets demonstrate the merit and viability of the proposed method compared with the current state-of-the-art methods.**

***Index Terms*— Endmember extraction, hyperspectral anomaly detection, image segmentation, low-rank prior, matrix decomposition.**

I. Introduction

HYPERSPECTRAL imaging allows a simultaneous observation of the geometrical and physical characteristics of materials [1]–[3], which makes it possible to precisely distinguish different objects of interest [4]. Target detection is one of the most important research fields of hyperspectral information processing. Hyperspectral target detection can generally be separated into anomaly detection and signature-based target detection [5]. In the anomaly detection task, no *a priori* knowledge is available about the anomalies or background [6], where anomalies refer to the observations with significant differences in spectral characteristic with the neighboring background [7].

The representation and suppression of the background are the significant issues in hyperspectral anomaly detection [8], [9]. To this end, the statistical methods utilize the prior statistical distributions for the background estimation. The well-known Reed–Xiaoli (RX) detector [10] assumes that the background obeys a multivariate normal distribution, and the background statistical variables are calculated in a local window, or based on the entire image, where anomaly contamination is unavoidable. Inspired by the RX detector (RXD), some improved methods have also been developed based on the multivariate normal distribution, including the regularized RXD [11], segmented RXD [12], weighted RXD, and linear filter-based RXD [13]. Among these methods, the segmented RXD applies the RXD in the segmentation regions after image segmentation, to solve the problem of anomaly contamination. Considering that a multivariate normal distribution is not sufficient to model the background classes, the random selection-based anomaly detector (RSAD) [14] assumes that the background obeys a chi-squared distribution and uses the subsets of the background pixels to compute the robust background statistics which present a better representation of the background. However, the common problem of the statistical methods is that the statistical distributions cannot truly describe the complicated background in a real hyperspectral image (HSI) [15].

To avoid the inappropriate statistical distribution assumptions, representation-based methods have been proposed, among which collaborative representation (CR) and sparse representation (SR) have drawn much attention. Li and Du [16] proposed a CR-based detector, which is based on the concept that each background pixel can be approximately linearly represented by its neighboring pixels, whereas the anomalies cannot. Furthermore, background pixels in a local region jointly exhibit such a characteristic, and based on this, background joint SR has been applied in hyperspectral anomaly detection [17]. Anomaly detection methods based on CR and SR can achieve good detection results; however,

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the background estimation process involves anomalous pixels, thus affecting the accuracy of the anomaly detection.

Recently, the low-rank prior [18]–[21] has been applied in hyperspectral anomaly detection to further improve the anomaly detection accuracy by separating the background from anomalies, based on the assumption that the background holds a low-rank property, whereas the anomalies exhibit a sparse property. The low-rank-based methods essentially transform the hyperspectral anomaly detection problem into a low-rank and sparse matrix decomposition (LRaSMD) problem. These methods can be generally divided into the robust principal component analysis (RPCA)-based methods [22], [23] and the low-rank representation (LRR)-based methods [15], [24], [25]. The RPCA-based methods assume that the pixel vectors in the HSI lie in a low-dimensional subspace and can be approximately represented as a linear combination of several basis vectors. This kind of method directly imposes the low-rank and sparse constraints on the original global data matrix, which is a 2-D matrix composed of all the pixel vectors in the HSI. An example of such a method is the LRaSMD-based Mahalanobis distance method for hyperspectral anomaly detection (LSMAD) [23]. However, the degree of anomaly–background separation is not satisfactory since exactly separating the background from anomalies is difficult [26]. The LRR-based methods utilize the low-rank property of the sparse coding coefficient matrix, in which the spectral dictionary is first constructed, and then, sparse coding is performed for the hyperspectral data, based on the dictionary. The low-rank and sparse constraints are then imposed on the sparse coefficient matrix, rather than the original global data matrix. The different methods of dictionary construction differentiate this kind of method. The anomaly detection method based on low-rank and SR (LRASR) [24] utilizes the pixels which have the smallest Mahalanobis distances in the k-means clusters as the atoms of the background dictionary. Similarly, in the abundance- and dictionary-based low-rank decomposition (ADLR) [25] method, centroid pixels and those with the longest distances to the centroid pixels in the mean-shift clusters of the abundance maps serve as the atoms of the dictionary. The LRR-based methods achieve a better separability between anomalies and background; however, the dictionary construction strategies are usually complicated, involving additional parameters.

In order to achieve a better anomaly–background separation while avoiding the need for complicated dictionary construction, a novel low-rank-based hyperspectral anomaly detection method based on a locally enhanced low-rank prior (LELRP-AD) is proposed in this article. This article is inspired by the observation that, in the local homogenous regions, the background signals hold an enhanced low-rank property, whereas the anomalies exhibit spatial sparsity, i.e., anomalous pixels account for only a small proportion of the total number of pixels [27]. Based on the above observation, background pixels can be low-rank reconstructed by a set of basis background signals [23], whereas anomalies are represented as sparse residuals, thus ensuring anomaly–background separation and realizing anomaly detection. However, two issues have to be addressed for the implementation: 1) generation of the local homogenous regions and 2) inducing the spatial sparsity of the anomalies. The local homogeneous regions should hold spatial and spectral continuity. Note that no prior information is available about the background or anomalies, so unsupervised image segmentation is adopted to generate the local homogeneous regions, where a Potts-based image segmentation method is utilized with postprocessing. In addition, the spatial sparsity of the anomalies is actually weakened in the local homogenous regions, as the ratio of anomalous pixels to the total number of pixels is increased compared with the entire image scene. Hence, the original local data matrix, which is composed of all the pixel vectors in one local homogenous region, is augmented with extracted background endmembers for the LRaSMD. Finally, the anomaly detection result is obtained by combining the sparse components in all the local homogeneous regions.

The main contributions of this article can be summarized as follows.

1) A novel hyperspectral anomaly detection method via LELRP-AD is proposed, which exploits the low-rank property of background signals in the local homogenous regions. Compared with the LRR-based methods, the proposed LELRP-AD method avoids the need for a complicated dictionary construction process.
2) The decomposition model is constructed in the local homogeneous regions, which are generated by a Potts-based image segmentation method with postprocessing. In these local homogeneous regions, the low-rank property of the background is enhanced, and the sparse background is further suppressed in the detection map.
3) The original local data matrix is augmented with the extracted background endmembers to induce the spatial sparsity of the anomalies. Constrained by the background spectral characteristics, the anomalous pixels are more accurately decomposed into the sparse component, thus ensuring anomaly–background separation.

The experimental results demonstrate that the proposed method shows a competitive anomaly detection performance, in both visualization and quantitative evaluation, compared with the state-of-the-art low-rank-based hyperspectral anomaly detection methods.

The rest of this article is organized as follows. Section II gives a brief introduction to the related work. Section III provides a detailed description of the proposed method. Section IV analyzes the experimental results and presents a parameter analysis for the proposed method. Finally, the conclusion is drawn in Section V.

II. Related Work

This section briefly summarizes the low-rank-based hyperspectral anomaly detection models, which can be generally divided into the RPCA-based methods and the LRR-based methods.

A. RPCA-Based Methods

Hyperspectral data have low intrinsic dimensionality [28], which means that the data lie in a low-dimensional subspace and are sparse on some basis [29]. To be specific, if all the

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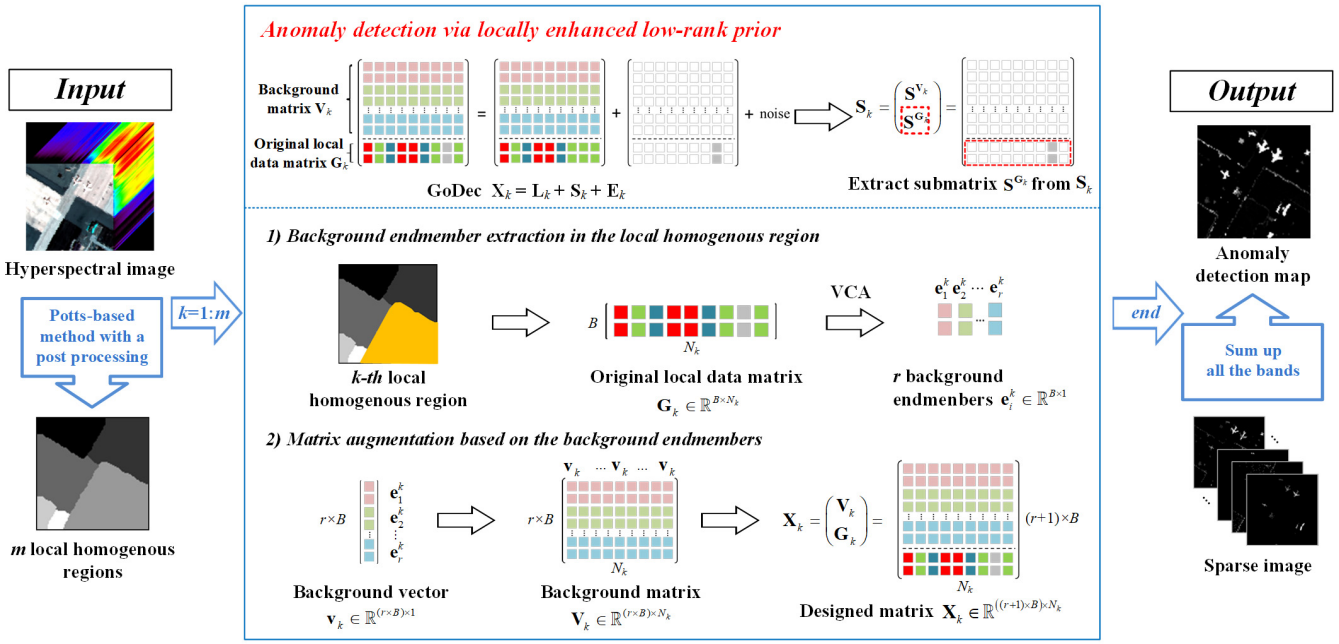


Fig. 1. Flowchart of the proposed LELRP-AD method.

pixel vectors in an HSI are stacked as columns of a matrix, most of the columns in this matrix can be approximately represented by the combination of several basis vectors, whereas those that cannot be represented by the basis vectors account for only a small proportion. It is thus reasonable to assume this matrix to be the superposition of a low-rank component $\mathbf{L}$, a sparse component $\mathbf{S}$, plus the noise component $\mathbf{E}$ [22], [23]

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{E}. \tag{1}$$

This model is simple and straightforward; however, exactly separating the background from anomalies is difficult, which affects the final anomaly detection accuracy.

B. LRR-Based Methods

To further separate the background from anomalies, the dictionary construction and sparse coding can be incorporated with RPCA [15], [24], [25]. In this way, the LRR-based methods decompose the hyperspectral data matrix $\mathbf{X}$ into a background part plus an anomalous part [24]

$$\mathbf{X} = \mathbf{DS} + \mathbf{E} \tag{2}$$

where $\mathbf{D}$ is the background dictionary which is composed of the background pixel vectors, and $\mathbf{S}$ is the representation coefficients. $\mathbf{DS}$ denotes the background part, and $\mathbf{E}$ denotes the anomalous part. It is assumed that $\mathbf{S}$ holds a low-rank property, whereas $\mathbf{E}$ has a sparsity.

Compared with the original image, the code image exhibits the purer representation of background, and thus, the LRR for the background is more accurate, achieving further background separation from anomalies compared with the RPCA-based methods.

On the basis of the above decomposition models, the regularization terms are introduced to calculate the low-rank property and sparsity. The nuclear norm is commonly used to calculate the low-rank property of the background. For the sparsity of the anomalies, the common regularization terms are the l_1-norm, $l_{2,1}$-norm, and $l_{1,2}$-norm.

It can be seen that the existing methods mainly model the global low-rank and sparse property, which is effective for the separation of background and anomalies; however, the complexity of the global background leads to uncertainty of the LRR for the background. Furthermore, the dictionary construction process usually involves the complicated strategies and additional parameters.

III. Hyperspectral Anomaly Detection via Locally Enhanced Low-Rank Prior

In order to further achieve anomaly–background separation while avoiding the need for a complicated dictionary construction strategy, a novel hyperspectral anomaly detection method via LELRP-AD is proposed in this article. The flowchart of the proposed LELRP-AD method is shown in Fig. 1, where the main steps are: 1) local homogenous region generation; 2) anomaly detection via the locally LELRP-AD; and 3) optimization for LELRP-AD. To better explain the proposed LELRP-AD method, a summary of the notation is shown in Table I.

A. Motivation

This article is motivated by the observation that, in local homogenous regions, the low-rank property of the background signals is enhanced while the spatial sparsity of the anomalies is maintained. Thus, the background pixels can be low-rank reconstructed by a set of basis background signals, whereas the anomalies are represented as sparse residuals. Based on different properties of the background and anomalies in

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TABLE I
NOTATION USED IN THIS ARTICLE

$\mathbf{X}=\{\mathbf{x}_i\}_{i=1}^{N}\in\mathbb{R}^{B\times N}$	**Original global data matrix**
$\mathbf{G}_k=\left[\mathbf{x}_1^k,\mathbf{x}_2^k,\ldots,\mathbf{x}_{N_k}^k\right]\in\mathbb{R}^{B\times N_k}$	Original local data matrix of the k-th local homogenous region
$\{\mathbf{e}_i^k\}_{i=1}^{r},\ \mathbf{e}_i^k\in\mathbb{R}^{B\times 1}$	Background endmembers of the k-th local homogenous region
$\mathbf{v}_k=\left[(\mathbf{e}_1^k)^T,(\mathbf{e}_2^k)^T,\ldots,(\mathbf{e}_r^k)^T\right]^T\in\mathbb{R}^{(r\times B)\times 1}$	Background vector of the k-th local homogenous region
$\mathbf{V}_k\in\mathbb{R}^{(r\times B)\times N_k}$	Background matrix of the k-th local homogenous region
$\mathbf{X}_k\in\mathbb{R}^{((r+1)\times B)\times N_k}$	Designed matrix of the k-th local homogenous region
$\mathbf{S}_k\in\mathbb{R}^{((r+1)\times B)\times N_k}$	Sparse component of the k-th local homogenous region
$\mathbf{S}^{\mathbf{G}_k}\in\mathbb{R}^{B\times N_k}$	Submatrix of $\mathbf{S}_k$ indicating the potential anomalous pixels
$\mathbf{S}\in\mathbb{R}^{B\times N}$	Sparse image

local homogenous regions, anomaly–background separation is ensured, and thus, anomaly detection is realized. Therefore, the novel hyperspectral anomaly detection method based on an LELRP-AD is proposed, which achieves further anomaly–background separation and avoids the need for a complicated dictionary construction process.

However, two issues have to be addressed for the implementation. The first issue is how to generate the local homogeneous regions, which should hold spatial and spectral continuity, given the fact that no prior information is available about the background or anomalies. To this aim, an unsupervised Potts-based image segmentation method is adopted [30] with postprocessing. Due to the reduced complexity of the background, there is no need for a dictionary to describe the background. Due to the homogeneity of the local regions, the variability of the background spectra is reduced, whereas the similarity is enhanced. Thus, the degree of linear fitting for the background pixel vectors is improved, i.e., the low-rank property of the background signals is enhanced. In addition, some background classes also account for a small proportion in regard to the total number of image pixels and exhibit spatial sparsity. In the local homogenous regions, these background classes become the dominant classes and the low-rank property becomes the dominant property, thus ensuring the separation from the sparse anomalies.

The second issue is how to induce the spatial sparsity of the anomalies in the local homogeneous regions. Note that the total number of pixels in a local region is decreased compared with the entire image, so the proportion of anomalous pixels is increased, i.e., the spatial sparsity of the anomalies is reduced. If the matrix decomposition is directly performed using the original local data matrix, the anomalous pixels may be decomposed into the low-rank component, in which case the response of the anomalous pixels will be decreased, and some anomalous pixels may not even be detected [31]. In order to enhance the spatial sparsity of the anomalies, the matrix is augmented for the LRaSMD, rather than directly using the original local data matrix. In this way, the newly designed matrix consists of two submatrices: the original local data matrix and an additional matrix. In order to maintain the low-rank property of the designed matrix, the additional matrix is defined as a rank-1 matrix. In order to ensure the correct decomposition, the additional matrix should be independent of the anomalies. Thus, the background endmembers are extracted based on which the additional matrix is constructed. Finally, the additional matrix is defined as a rank-1 matrix that represents the background characteristics, which is called the background matrix in this article.

The spatial sparsity of the anomalies is enhanced in the designed matrix. For the implementation of LRaSMD, the spatial sparsity refers to the elements corresponding to anomalies being nonzero, whereas the other elements are 0 in the sparse component, and the nonzero elements only account for a small proportion of the total elements. In the designed matrix, the additional matrix is decomposed into the low-rank component since it is a rank-1 matrix. Thus, the nonzero elements in the sparse component will only come from the original local data matrix. That is to say, the number of nonzero elements remains unchanged, whereas the total number of matrix elements is increased after the matrix augmentation. In this way, the proportion of nonzero elements in the designed matrix is decreased, i.e., the spatial sparsity of the anomalies is enhanced.

B. Local Homogenous Region Generation

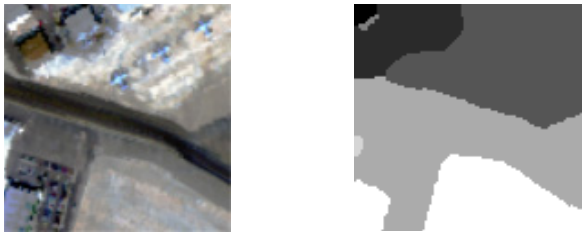
In order to generate the local homogeneous regions, unsupervised image segmentation is utilized, where a Potts-based method [30] is adopted with postprocessing.

Based on the assumption that the grayscale of the image changes slowly within a homogenous region but changes sharply on the boundary between two homogenous regions, the Potts-based method is given by the following minimization problem:

$$\mathbf{U}^*=\arg\min_{\mathbf{U}}\ \rho\,\|\nabla\mathbf{U}\|_0+\|\mathbf{U}-\mathbf{X}_{\mathrm{PCA}}\|_2^2 \tag{3}$$

where $\|\nabla\mathbf{U}\|_0$, which is called the boundary term, denotes the length of the boundaries of the segmentation regions since the boundaries of the segmentation regions agree with the support of the gradient $\nabla\mathbf{U}$ [30]; $\|\mathbf{U}-\mathbf{X}_{\mathrm{PCA}}\|_2^2$ is the data fidelity term; ρ is the scale parameter adjusting the balance between the two terms, where a large value of ρ leads to large segmentation regions and vice versa; $\mathbf{X}_{\mathrm{PCA}}\in\mathbb{R}^{3\times N}$ is the stack of the first three principal components [32] of the original hyperspectral data, through which the redundant information in the hyperspectral data is avoided; and $\mathbf{U}$ is the optimized image with the same dimension as $\mathbf{X}_{\mathrm{PCA}}$. A three-band probability map is obtained after the optimization of (3), and then, the same label is given to the equal probability vectors to obtain the segmentation map.

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(a) (b)

Fig. 2. Illustration of the image segmentation. (a) RGB pseudocolor image. (b) Local homogenous regions.

The boundary term is defined on an eight-connected neighborhood $U = \{(1, 0), (0, 1), (-1, 1), (1, 1)\}$, specifically

$$\|\nabla \mathbf{U}\|_0 = \sum_{i,j} \sum_{(a,b)\in U} w_{ab}[\mathbf{u}_{i,j,:} \neq \mathbf{u}_{i+a,j+b,:}] \tag{4}$$

where w_{ab} is the constant neighborhood weights, in which $w_{1,0} = w_{0,1} = \sqrt{2}-1$, $w_{-1,1} = w_{1,1} = 1-(\sqrt{2}/2)$, and $\mathbf{u}_{i,j}$, is the vector sitting at pixel (i, j), for which the value in the bracket $[\cdot]$ is 1, if the expression inside is true, and 0 otherwise. The optimization problem (3) can be solved by the alternating direction method of multipliers (ADMM) approach [30].

After the Potts-based method generates the regions, a post-processing is performed. Note that the anomalies are usually small, and a region in which the total number of pixels is less than 20 pixels is merged into its adjacent region with the highest spectral similarity, which is defined as the minimum spectral angle distance between the mean vectors of adjacent regions. After the image segmentation, the HSI is segmented into m local homogeneous regions, which can be represented as follows:

$$\mathbf{X} = \cup_{k=1,\ldots,m}\mathbf{G}_k,\ \mathbf{G}_k \cap \mathbf{G}_i = \varnothing,\quad k \neq i,\ k, i = 1, \ldots, m \tag{5}$$

where $\mathbf{G}_k$ is the original local data matrix that contains the pixel vectors in the kth local homogenous region, i.e., $\mathbf{G}_k = [\mathbf{x}_1^k, \mathbf{x}_2^k, \ldots, \mathbf{x}_{N_k}^k] \in \mathbb{R}^{B\times N_k}$, and N_k represents the total number of pixels in the kth segmentation region.

The segmentation follows the principle of roughly segmenting the dominant background classes into independent regions, to ensure the homogeneity of the local regions, as shown in Fig. 2. Furthermore, oversegmentation is avoided since the anomalies tend to be segmented into independent regions under oversegmentation. The Potts-based segmentation method controls the degree of segmentation by adjusting the weight of the boundary term, and thus, the segmentation boundaries tend to be along the borders of the dominant background classes. In addition, it is easy to avoid oversegmentation with the Potts-based method by raising the weight of the boundary term, since the scale parameter ρ varies in the range of (0, 1), and has a clear correlation with the degree of segmentation.

C. Anomaly Detection via Locally Enhanced Low-Rank Prior

Now, the local homogenous regions have been generated, and the data in each region are then decomposed via LELRP-AD, in which LRaSMD theory is utilized. Since the procedures are the same in all the segmentation regions, the descriptions of the proposed method in this section are based on the kth segmentation region as an example. In the proposed method, the data in each region are assumed to be composed of a low-rank component, a sparse component, plus a noise component

$$\mathbf{X}_k = \mathbf{L}_k + \mathbf{S}_k + \mathbf{E}_k \tag{6}$$

where a novel strategy is designed in this article for the decomposition, i.e., a novel matrix $\mathbf{X}_k$ is constructed to induce the spatial sparsity of the anomalies. $\mathbf{L}_k$, $\mathbf{S}_k$, and $\mathbf{E}_k$ are the low-rank component, sparse component, and noise component, respectively. As mentioned above, the designed matrix consists of two submatrices, which are the original local data matrix and the background matrix. The original local data matrix is composed of the pixel vectors in the region. Therefore, the main issue is to construct a rank-1 background matrix.

First, background estimation is necessary for each local homogenous region. To this aim, the vertex component analysis (VCA) endmember extraction method [33] is utilized, which is both stable and simple. The number of endmembers has to be set in VCA, which is determined according to the number of dominant background classes. Note that the local regions are homogeneous, so the number of endmembers should be small. An empirical range for this parameter is given in the following parameter analysis section. Although the background classes are different within each region, all the local regions are set to the same number of endmembers to reduce the number of parameters.

It is assumed that the extracted endmembers mainly represent the background spectral characteristics. Through VCA, background endmembers of the kth region are extracted, i.e., $\{\mathbf{e}_i^k\}_{i=1}^r$, $\mathbf{e}_i^k \in \mathbb{R}^{B\times 1}$, where r is the number of endmembers. All the endmembers are then stacked to form a longer vector using the following equation, which is called the "background vector" and represents the background spectral characteristics:

$$\mathbf{v}_k = \left[(\mathbf{e}_1^k)^{\mathrm{T}}, (\mathbf{e}_2^k)^{\mathrm{T}}, \ldots, (\mathbf{e}_r^k)^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{(r\times B)\times 1}. \tag{7}$$

The background vector is then arranged in columns to form the background matrix $\mathbf{V}_k$

$$\mathbf{V}_k = [\mathbf{v}_k, \mathbf{v}_k, \ldots, \mathbf{v}_k] \in \mathbb{R}^{(r\times B)\times N_k} \tag{8}$$

where the number of columns of $\mathbf{V}_k$ is equal to the total number of pixels in the kth region, and the number of rows is equal to the length of the background vector. Finally, the designed matrix in the proposed method is composed of two parts: 1) the background matrix, which represents the background spectral characteristics; and 2) the original local data matrix $\mathbf{G}_k$, which contains the pixel vectors in the kth local homogenous region

$$\mathbf{X}_k = \begin{pmatrix} \mathbf{V}_k \\ \mathbf{G}_k \end{pmatrix} \in \mathbb{R}^{((r+1)\times B)\times N_k}. \tag{9}$$

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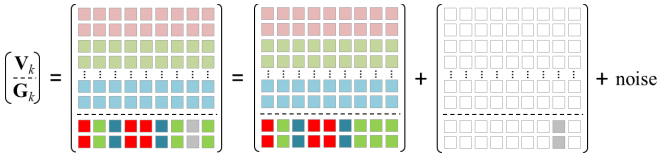


Fig. 3. Illustration of the LELRP-AD.

If the LRaSMD is directly performed in each region, the matrix will only be $\mathbf{G}_k$. The difference of the designed matrix is the addition of the background matrix $\mathbf{V}_k$. Clearly, $\mathbf{V}_k$ is a rank-1 matrix. If we suppose that anomalous pixels exist in $\mathbf{G}_k$ and the total number is A_k, then the total number of nonzero elements is $A_k \times B$. The ratio of nonzero elements in the original local data matrix $\mathbf{G}_k$ is then $(A_k \times B)/(B \times N_k)$, whereas the ratio in the designed matrix $\mathbf{X}_k$ is $(A_k \times B) / ((r+1) \times B \times N_k)$. The number of nonzero elements remained unchanged, whereas the total number of matrix elements is increased after the matrix augmentation, so that the proportion of nonzero elements in the designed matrix is decreased. In this way, the spatial sparsity of the anomalies is enhanced.

In the following, the relationships between the rank of $\mathbf{X}_k$, $\mathbf{V}_k$, and $\mathbf{G}_k$ are discussed. First, the rank of the matrix has the following relationship with its dimension:

$$\begin{aligned} 0 &\le \operatorname{rank}(\mathbf{V}_k) \le \min(r \times B, N_k) \\ 0 &\le \operatorname{rank}(\mathbf{G}_k) \le \min(B, N_k) \\ 0 &\le \operatorname{rank}(\mathbf{X}_k) \le \min\left((r+1) \times B, N_k\right). \end{aligned} \tag{10}$$

The designed matrix can actually be regarded as the sum of the two submatrices, as shown in the following equation:

$$\operatorname{rank}(\mathbf{X}_k) \le \operatorname{rank}\begin{pmatrix} 0 \\ \mathbf{G}_k \end{pmatrix} + \operatorname{rank}\begin{pmatrix} \mathbf{V}_k \\ 0 \end{pmatrix} \tag{11}$$

where $\operatorname{rank}\begin{pmatrix} 0 \\ \mathbf{G}_k \end{pmatrix} = \operatorname{rank}(\mathbf{G}_k)$, $\operatorname{rank}\begin{pmatrix} \mathbf{V}_k \\ 0 \end{pmatrix} = 1$, and thus the rank of $\mathbf{X}_k$ has the following relationship with the rank of $\mathbf{G}_k$, which also explains why $\mathbf{X}_k$ is a low-rank matrix:

$$\operatorname{rank}(\mathbf{X}_k) \le \operatorname{rank}(\mathbf{G}_k) + 1. \tag{12}$$

Fig. 3 shows an illustration of a decomposition model based on the LELRP-AD. If a pixel vector, which is a column of $\mathbf{G}_k$, belongs to the background, it is similar to the column of $\mathbf{V}_k$ and is decomposed into the low-rank component. If a pixel vector belongs to the anomalies, it is decomposed into the sparse component. In Fig. 3, the white squares represent 0 values and the gray squares represent nonzero values. In the sparse component, most of the elements are 0. The columns which have the nonzero elements indicate anomaly contamination.

D. Optimization for LELRP-AD

To separate the anomalies from the background and noise, the GoDec algorithm [34] is utilized to solve the problem in (6), i.e., decomposing the designed matrix $\mathbf{X}_k$ into a low-rank matrix $\mathbf{L}_k$, a sparse matrix $\mathbf{S}_k$, plus the noise $\mathbf{E}_k$, where $\mathbf{S}_k$ indicates the potential anomalies in the local homogenous region.

The approximated decomposition problem in (6) can be solved by solving the optimization problem in the following equation:

$$\begin{aligned} &\min_{\mathbf{L}_k, \mathbf{S}_k} \|\mathbf{X}_k - \mathbf{L}_k - \mathbf{S}_k\|_F^2 \\ &\text{s.t. } \operatorname{rank}(\mathbf{L}_k) \le r \\ &\quad \operatorname{card}(\mathbf{S}_k) \le c \times ((r+1) \times B \times N_k) \end{aligned} \tag{13}$$

where r is the upper bound of the rank of $\mathbf{L}_k$, and $c \times ((r+1) \times B \times N_k)$ is the upper bound of the cardinality of $\mathbf{S}_k$, which reflects the sparse energy in the matrix. In this article, the cardinality is defined as the l_0-norm of $\mathbf{S}_k$. Note that the total number of elements in $\mathbf{S}_k$, i.e., $(r+1) \times B \times N_k$ is a constant, and we only consider the impact of the ratio parameter c on the results. The value of $r$is highly correlated with the number of dominant background endmembers or background classes [35]. Therefore, the rank value is set as the same as the number of endmembers in VCA and uniformly represented as r.

The optimization problem in (13) can be solved using the GoDec algorithm by alternately solving the subproblems in (14). This is an iterative process until convergence, i.e., $\left\|\mathbf{X}_k - L_k^t - \mathbf{S}_k^t\right\|_F^2 / \|\mathbf{X}_k\|_F^2 < \varepsilon$, where ε is the threshold, which is set to 10^{-7}, and t refers to the tth iteration. Actually, the maximum number of iterations is set to 5 for the implementation. The matrix $\mathbf{L}_k$ is initialized as the designed matrix $\mathbf{X}_k$, and the matrix $\mathbf{S}_k$ is initialized as the null matrix, which has the same dimensions as $\mathbf{L}_k$

$$\begin{cases} \mathbf{L}_k^t = \underset{\operatorname{rank}(\mathbf{L}_k) \le r}{\arg\min} \left\|\mathbf{X}_k - \mathbf{L}_k - \mathbf{S}_k^{t-1}\right\|_F^2 \\ \mathbf{S}_k^t = \underset{\operatorname{card}(\mathbf{S}_k) \le cN_k}{\arg\min} \left\|\mathbf{X}_k - \mathbf{L}_k^t - \mathbf{S}_k\right\|_F^2. \end{cases} \tag{14}$$

The solution of (14) is defined in the following equations:

$$\mathbf{L}_k^t = Y_1 \left(A_2^{\mathrm{T}} Y_1\right)^{-1} Y_2^{\mathrm{T}} \tag{15}$$

$$\begin{aligned} &\mathbf{S}_k^t = P_\Omega\left(\mathbf{X}_k - \mathbf{L}_k^t\right), \ \Omega: \ \left|\left(\mathbf{X}_k - \mathbf{L}_k^t\right)_{i,j \in \Omega}\right| \ne 0 \\ &\text{and} \ \ge \left|\left(\mathbf{X}_k - \mathbf{L}_k^t\right)_{i,j \in \bar{\Omega}}\right|, \quad |\Omega| \le cN_k \end{aligned} \tag{16}$$

where $A_1 \in \mathbb{R}^{N_k \times r}$ and $\mathbf{A}_2 \in \mathbb{R}^{B \times r}$ are the random matrices, and $Y_1 = (\mathbf{X}_k - \mathbf{S}_k^{t-1})\mathbf{A}_1$, $Y_2 = (\mathbf{X}_k - \mathbf{S}_k^{t-1})^{\mathrm{T}} Y_1$. $\mathbf{S}_k^t$ is updated via the entry-wise hard thresholding of $\mathbf{X}_k - \mathbf{L}_k^t$ [36]. Ω is the nonzero subset of the first $c \times ((r+1) \times B \times N_k)$ largest entries of $|\mathbf{X}_k - \mathbf{L}_k^t|$, and $\bar{\Omega}$ is the corresponding complementary set.

The sparse matrix $\mathbf{S}_k$ is obtained after the GoDec-based LRaSMD. $\mathbf{S}_k$ is actually composed of two parts, one from $\mathbf{V}_k$ and the other from $\mathbf{G}_k$

$$\mathbf{S}_k = \begin{pmatrix} \mathbf{S}^{\mathbf{V}_k} \\ \mathbf{S}^{\mathbf{G}_k} \end{pmatrix} \in \mathbb{R}^{((r+1) \times B) \times N_k} \tag{17}$$

where $\mathbf{S}^{\mathbf{G}_k} \in \mathbb{R}^{B \times N_k}$ is a submatrix of $\mathbf{S}_k$ that indicates the potential anomalous pixels. Each column of $\mathbf{S}^{\mathbf{G}_k}$ corresponds to one pixel of the kth region. The sparse image $\mathbf{S} \in \mathbb{R}^{B \times N}$

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is recovered after all the regions are finished

$$\mathbf{S} = \cup_{k=1,\ldots,m}\mathbf{S}^{\mathbf{G}_k}, \mathbf{S}^{\mathbf{G}_k} \cap \mathbf{S}^{\mathbf{G}_i} = \varnothing, \quad k \neq i,\ k, i = 1, \ldots, m. \quad (18)$$

For the output of the detection map, all the bands of the sparse image are summed using the following equation, where $\mathbf{s}_i$ is the row vector of $\mathbf{S}$:

$$D(\mathbf{S}) = \sum_{i=1}^{B} \mathbf{s}_i. \quad (19)$$

The proposed method is summarized in Algorithm 1.

Algorithm 1 The Proposed LELRP-AD Method

Input:

$\mathbf{X} \in \mathbb{R}^{B\times N}$

Initialization:

Parameters: ρ, r, c

1: Perform image segmentation on $\mathbf{X}$ under the scale parameter ρ

$$\mathbf{X} = \cup_{k=1,\ldots,m}\mathbf{G}_k, \mathbf{G}_k \cap \mathbf{G}_i = \varnothing, \quad k \neq i,\ k, i = 1, \ldots, m$$

2: for $k = 1{:}m$

2.1: Extract the background endmembers $\left\{\mathbf{e}_i^k\right\}_{i=1}^{r}, \mathbf{e}_i^k \in \mathbb{R}^{B\times 1}$ of $\mathbf{G}_k$

2.2: Construct the background matrix $\mathbf{V}_k$ using (7)–(8)

2.3: Construct the designed matrix $\mathbf{X}_k$ using (9)

2.4: Decompose $\mathbf{X}_k$ into a sparse component $\mathbf{S}_k$, a low-rank component $\mathbf{L}_k$, and a noise component $\mathbf{E}_k$ by solving the optimization problem in (13)

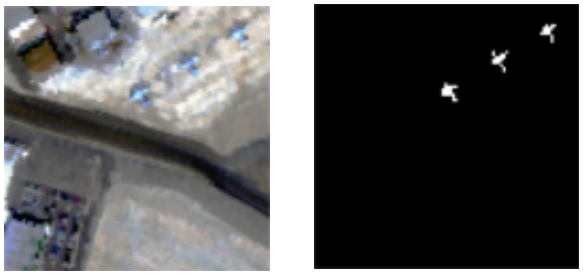
2.5: Extract the submatrix $S^{\mathbf{G}_k}$ from $\mathbf{S}_k$

3: Obtain the sparse image $\mathbf{S} = \cup_{k=1,\ldots,m}\mathbf{S}^{\mathbf{G}_k}$, $\mathbf{S}^{\mathbf{G}_k} \cap \mathbf{S}^{\mathbf{G}_i} = \varnothing,\ k \neq i,\ k, i = 1, \ldots, m$

4: Anomaly detection $D(\mathbf{S}) = \sum_{i=1}^{B} \mathbf{s}_i$

Output:

The anomaly detection map



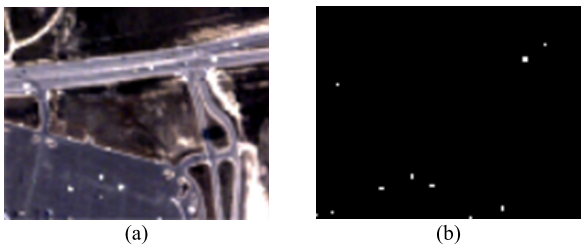
(a) (b)

Fig. 4. AVIRIS-1 data set. (a) RGB pseudocolor image. (b) Ground-truth map.



(a) (b)

Fig. 5. AVIRIS-2 data set. (a) RGB pseudocolor image. (b) Ground-truth map.



(a) (b)

Fig. 6. HYDICE data set. (a) RGB pseudocolor image. (b) Ground-truth map.

IV. Experiments and Analyses

A. Experimental Settings

1) Data Set Description: Four hyperspectral data sets were used in the experiments. The first two data sets were the subsets of images collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor [37] in different areas of San Diego Airport, CA, USA. The spectral range of the imagery is 370–2510 nm, and the spatial resolution is 3.5 m per pixel. The image scene is of 100 × 100 pixels in size. For the AVIRIS-1 data set, the number of spectral bands is 189, and three planes are considered as the anomalies, with a total number of 58 pixels, as shown in Fig. 4(a) and (b). For the AVIRIS-2 data set, the number of spectral bands is 186, and three planes are again considered as the anomalies, with a total number of 143 pixels, as shown in Fig. 5(a) and (b).

The third data set was collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor [38] over a suburban residential area of Michigan, USA. The spectral range of the imagery is 400–2500 nm, and the spatial resolution is 3 m per pixel. The image scene is of 80 × 100 pixels in size. In total, 162 spectral bands were retained after removing the bands of the water absorption regions, low signal-to-noise ratio, and poor quality (1–4, 76, 87, 101–111, 136–153, and 198–210). Ten man-made vehicles are considered as the anomalies, with a total number of 17 pixels, and the background land-cover types include parking lot, soil, water, and road, as shown in Fig. 6(a) and (b).

The fourth data set was the Honghu data set collected by the Intelligent Data Extraction, Analysis, and Application of Remote Sensing (RSIDEA) Group of Wuhan University, Wuhan, China [39]. This data set was obtained by a Nano-Hyperspec visible and near-infrared UAV-borne hyperspectral sensor. The data collection was carried out on March 21, 2018. The image scene is a long river bank in Honghu, Hubei Province of China. The spatial resolution of the imagery is 6 cm per pixel, and the image scene is of 105 × 168 pixels in size, as shown in Fig. 7(a). The image has 135 bands, and the spectral range is 400–1000 nm. Two plastic plates are

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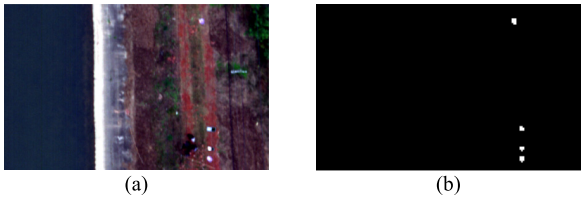


Fig. 7. Honghu data set. (a) RGB pseudocolor image. (b) Ground-truth map.

considered as the anomalies, with a total number of 21 pixels, as shown in Fig. 7(b).

2) Parameter Settings: The compared algorithms were the classical RXD [10], segmentation-based RXD (SegRX) [12], and three state-of-the-art low-rank-based hyperspectral anomaly detection methods: 1) the anomaly detection method based on LRASR [24]; 2) the LRaSMD-based Mahalanobis distance method for hyperspectral anomaly detection (LSMAD) [23]; and 3) the ADLR method for hyperspectral anomaly detection [25].

The SegRX method used the same segmentation map as the proposed method, to make the comparison fair. According to the description in this article of LRASR, the number of clusters K and the number of selected pixels P were set to 15 and 20, respectively, and the regularization parameters β and λ were set to 0.1 for all the experimental data sets.

The cardinality k and the rank value r in LSMAD were set to 0.004 and 2 for the AVIRIS-1 data set, respectively, according to the recommendations of the authors. Note that k has little effect on the detection accuracy under the same rank value. To ensure that all the anomalous pixels could be detected, k was set to 0.004 for the other three data sets. According to the description in the original article of LSMAD, an empirical range of rank value is [1, 2], depending on the number of dominant categories in the HSI. Therefore, r was set to 2 for the other three data sets.

The ADLR method has three free parameters: the number of endmembers c, the bandwidth in the mean-shift algorithm bw, and the regularization parameter λ. We conducted a large number of experiments to obtain the best performance for every data set. Finally, the parameters were set as follows: 1) for the AVIRIS-1 data set, we used $c = 7$, bw $= 0.2$, and $\lambda = 0.02$; 2) for the AVIRIS-2 data set, we used $c = 5$, bw $= 0.28$, and $\lambda = 0.02$; 3) for the HYDICE data set, we used $c = 21$, bw $= 0.21$, and $\lambda = 0.02$; and 4) for the Honghu data set, we used $c = 10$, bw $= 0.2$, and $\lambda = 0.02$.

For the proposed method, the scale parameter ρ was set to 0.25, 0.5, 0.45, and 0.11 for the AVIRIS-1, AVIRIS-2, HYDICE, and Honghu data sets, respectively. The cardinality c was set to 0.03 for the AVIRIS-2 data set. For the other three data sets, the cardinality c was set to 0.01. The number of endmembers is equal to the rank value. These two parameters are referred to as r in this article. The value of r was determined empirically, according to the number of dominant background classes in the local homogenous regions. For the AVIRIS-1 and AVIRIS-2 data sets, r was set to 2; for the HYDICE and Honghu data sets, r was set to 5 and 1, respectively.

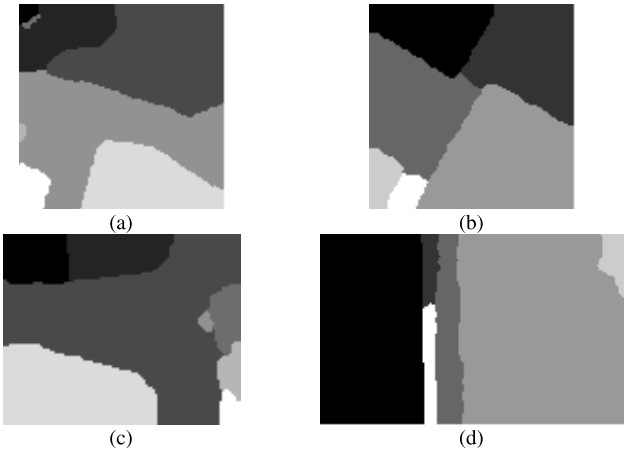


Fig. 8. Segmentation maps of the experimental data sets. (a) AVIRIS-1. (b) AVIRIS-2. (c) HYDICE. (d) Honghu.

B. Experimental Results

In this section, we qualitatively compare the detection results of the proposed method with those of the other competing methods. The segmentation maps are shown in Fig. 8, where the dominant background classes are segmented into independent regions. The scale parameter is set to avoid over-segmentation where the anomalies are contained in the background regions and not segmented into independent regions since the anomalies are rather small. The 2-D plots for the obtained detection results are shown in Figs. 9–12. All the detection maps are linearly stretched to [0, 255] for display. For the AVIRIS-1 data set, all the experimental methods can detect the anomalies, except for RXD, where the middle plane is barely visible and the other two planes show a weak response. For the AVIRIS-2 data set, the proposed method exhibits a higher contrast between anomalies and background than the other methods. For the HYDICE data set, the anomalies are not visible in the results of LSMAD and SegRX. We can see that the background is better suppressed in the result of the proposed method compared with the other methods. For the Honghu data set, there are no major false alarms in the result of the proposed method. From a visual comparison, the proposed method provides a higher discrimination between anomalies and background on all the experimental data sets.

The effectiveness of the proposed LELRP-AD method in suppressing the sparse background is now demonstrated. For the AVIRIS-1 data set, the sparse background mainly refers to the area in the lower left quarter of the image scene, which is marked with a red box for display. It can be seen that two stripes with a strong response remain in the results of RXD. A rectangular region with strong response remains in the results of LRASR, LSMAD, and ADLR. These sparse background regions are effectively suppressed in the result of the proposed method. For the AVIRIS-2 data set, a rectangular background region with a strong response remains in the results of RXD and LRASR. The roads in the lower right corner are well marked in the result of SegRX. The anomalies are all well marked in the results of LSMAD, ADLR,

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Fig. 9. Target detection results on the AVIRIS-1 data set. (a) RXD. (b) SegRX. (c) LRASR. (d) LSMAD. (e) ADLR. (f) LELRP-AD.

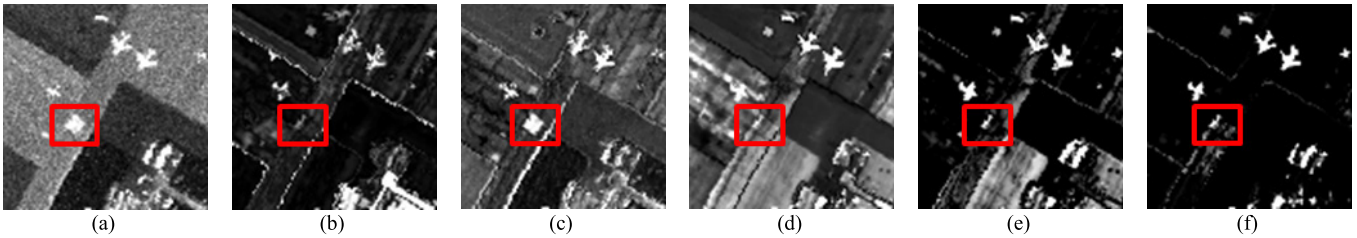


Fig. 10. Target detection results on the AVIRIS-2 data set. (a) RXD. (b) SegRX. (c) LRASR. (d) LSMAD. (e) ADLR. (f) LELRP-AD.

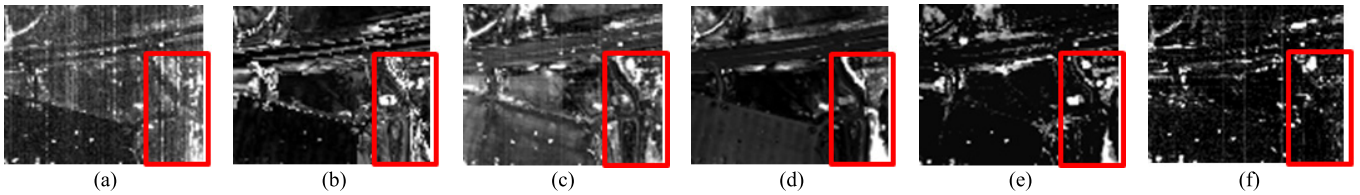


Fig. 11. Target detection results on the HYDICE data set. (a) RXD. (b) SegRX. (c) LRASR. (d) LSMAD. (e) ADLR. (f) LELRP-AD.

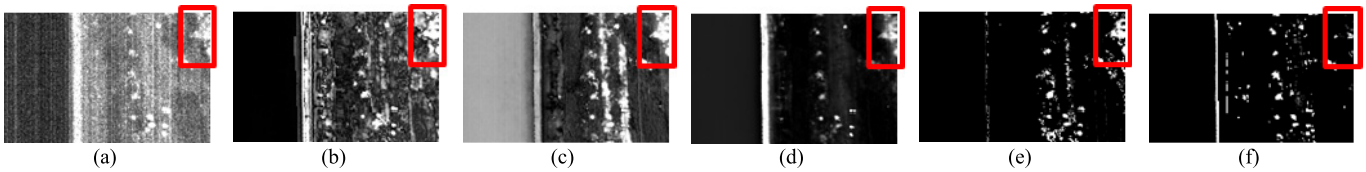


Fig. 12. Target detection results on the Honghu data set. (a) RXD. (b) SegRX. (c) LRASR. (d) LSMAD. (e) ADLR. (f) LELRP-AD.

and proposed method, among which the proposed method achieves an overall lower response for the background. For the HYDICE data set, the road in the lower right corner of the image scene stands out in the results of LSMAD, which is suppressed, to some extent, due to the dictionary construction and sparse coding process in LRASR and ADLR, but not sufficiently. On the contrary, the road is barely visible in the result of the proposed method. For the Honghu data set, the vegetation in the upper right corner represents a strong response in the results of RXD, SegRX, LRASR, LSMAD, and ADLR. On the contrary, the proposed method effectively suppresses these background regions, achieving a better anomaly–background separation. From the above analysis, we can see that the proposed method demonstrates a much better background–anomaly separation and is, especially, effective for the suppression of the sparse background.

C. Quantitative Evaluation

To quantitatively evaluate the results, receiver operating characteristic (ROC) [40] curves and the area under the ROC curve (AUC) [41] are employed. The ROC curves for these experiments are shown in Fig. 13(a)–(d). The corresponding AUC values are shown in Table II, where the figures in bold are the best results and the underlined figures are the second-best results.

TABLE II
AUC SCORES FOR DIFFERENT METHODS ON THE EXPERIMENTAL DATA SETS

	RXD	SegRX	LRASR	LSMAD	ADLR	Proposed
AVIRIS-1	0.8795	0.9706	0.9864	<u>0.9906</u>	0.9886	**0.9916**
AVIRIS-2	0.9370	0.9482	0.9146	0.8989	<u>0.9613</u>	**0.9879**
HYDICE	<u>0.9938</u>	0.8585	0.9937	0.6727	0.9624	**0.9950**
Honghu	**0.9989**	0.9929	0.9879	0.9542	0.9313	<u>0.9983</u>

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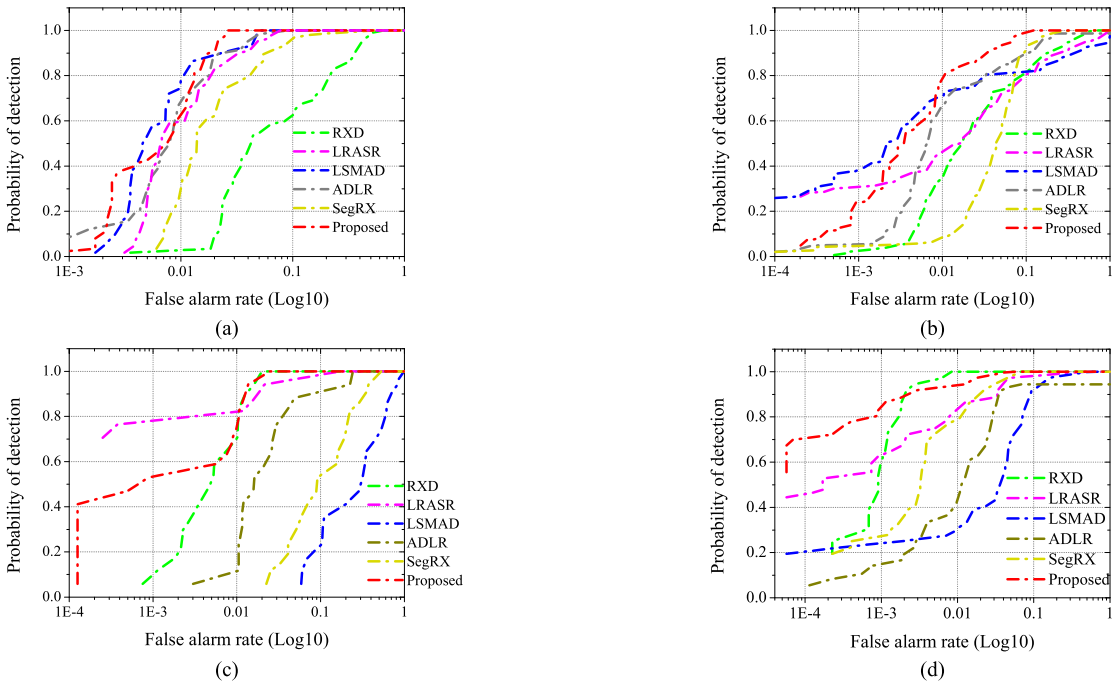


Fig. 13. ROC curves of the hyperspectral data sets. (a) AVIRIS-1 data set. (b) AVIRIS-2 data set. (c) HYDICE data set. (d) Honghu data set.

For the AVIRIS-1 data set, the ROC curve of the proposed LELRP-AD method is almost always above those of RXD, LRASR, ADLR, and SegRX while it intersects with that of LSMAD. The ROC curve of the proposed method is above that of LSMAD when the false alarm rate (FAR) is 0–0.004 and 0.017–1. When achieving a 100% probability of detection, the proposed method and LSMAD have FARs of approximately 0.026 and 0.05, respectively. Overall, the proposed method achieves the highest AUC score among all the methods.

For the AVIRIS-2 data set, the ROC curve of the proposed LELRP-AD method is always above those of RXD, SegRX, and ADLR while it intersects with those of LRASR and LSMAD. The ROC curve of the proposed method is slightly below that of LSMAD when the FAR is 0.0001–0.0095 and below that of LRASR when the FAR is 0.0001–0.0028. However, the proposed method exhibits a higher probability of detection than LSMAD and LRASR after the FAR increases to 0.0095 and 0.0028, respectively. When achieving a 100% probability of detection, the proposed method, LRASR, and LSMAD have FARs of approximately 0.12, 0.98, and 1, respectively. The proposed method achieves the highest overall accuracy, with the highest AUC score of 0.9889, which is 0.0276 higher than the second-best result of ADLR.

The proposed LELRP-AD method also shows a good detection performance on the HYDICE and Honghu data sets, where the anomalies are small in size.

For the HYDICE data set, the ROC curve of the proposed LELRP-AD method is always above those of RXD, LSMAD, ADLR, and SegRX. Although the proposed method exhibits a lower probability of detection than LRASR for a low FAR (when the FAR is 0.0006–0.01), the ROC curve is always above that of LRASR after the FAR increases beyond 0.01. The proposed method also achieves the highest AUC score of 0.9950.

For the Honghu data set, the ROC curve of the proposed LELRP-AD method is almost always above those of LRASR, LSMAD, ADLR, and SegRX while it intersects with that of RXD. We can also see that the proposed method exhibits a higher probability of detection than RXD for a low FAR (when the FAR is 0–0.0018). When the FAR increases to 0.0018, the probability of detection is 0.86. Overall, the proposed method achieves a better detection performance than the state-of-the-art low-rank-based methods on this data set.

The proposed LELRP-AD method can be considered as the most robust when compared with the other competing methods, including the state-of-the-art low-rank-based methods. The line charts of the AUC scores of the different methods on all four data sets are shown in Fig. 14. In Fig. 14, it can be seen that the proposed method exhibits the least variation in AUC score among all the competing methods. ADLR also achieves a robust detection performance, whereas the AUC scores are less than those of the proposed method. The other methods all show significant precision variations on different data sets, among which LSMAD presents the maximum variation in accuracy.

D. Discussion

In this section, the following four issues are discussed: 1) the effectiveness of the designed matrix; 2) the ability of the proposed method to effectively suppress the sparse background; 3) a demonstration of the idea of the "enhanced low-rank prior;" and 4) the computational complexity of the proposed method.

In order to verify the effectiveness of the designed matrix, the detection results based on the designed matrix and those

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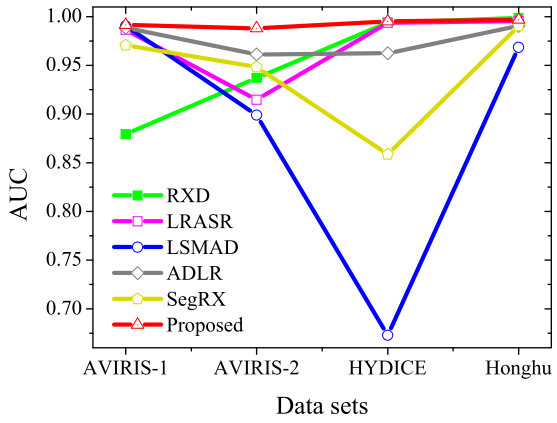


Fig. 14. Line charts of the AUC scores.

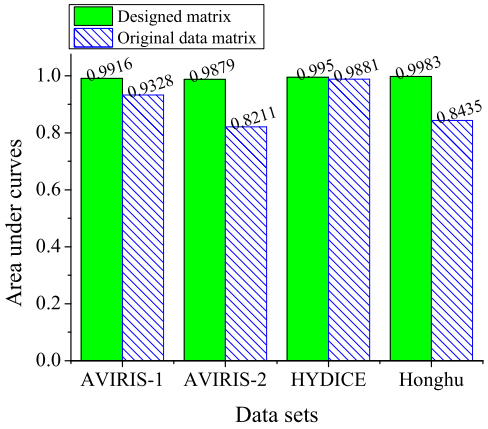


Fig. 15. Comparison between the results with the designed matrix and the results with the original local data matrix.

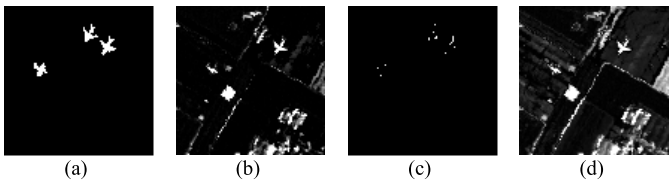


Fig. 16. Illustration of the effectiveness of the designed matrix with the AVIRIS-2 data set. (a) Ground-truth map. (b) Detection map with the original local data matrix under $c = 0.03$. (c) Undetected anomalous pixels. (d) Detection map with the original local data matrix under $c = 0.13$.

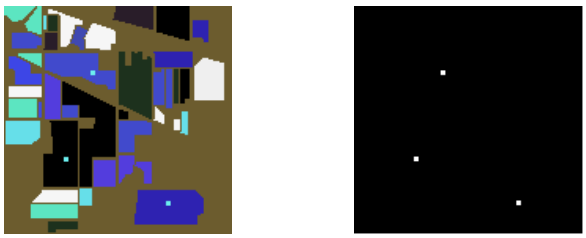


Fig. 17. Simulated data set. (a) RGB pseudocolor image. (b) Ground-truth map.

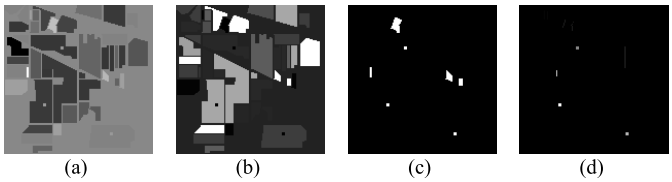


Fig. 18. Detection results for the simulated data set. (a) LRASR. (b) LSMAD. (c) ADLR. (d) LELRP-AD.

directly based on the original local data matrix are compared. All the parameters were set the same for both the designed matrix and the original data matrix, i.e., $\rho = 0.5$, $r = 2$, and $c = 0.03$. For all the experimental data sets, as shown in Fig. 15, the AUC score based on the designed matrix is always higher. Except for the HYDICE data set, some anomalous pixels are undetected with the original local data matrix under the same cardinality, which indicates that the designed matrix requires a smaller cardinality to detect all the anomalous pixels. Therefore, the designed matrix is more capable of inducing the spatial sparsity of the anomalies.

We take the AVIRIS-2 data set as an example. If the LRaSMD is directly performed using the original local data matrix, some anomalous pixels cannot be detected under the cardinality of $c = 0.03$, whereas the anomalous pixels are all detected in the result based on the designed matrix. The corresponding detection map is shown in Fig. 16(b), and the map of the undetected anomalous pixels is shown in Fig. 16(c). Under the circumstances, we increased the value of c. Finally, all the anomalous pixels are detected under $c = 0.13$, as shown in Fig. 16(d). The detection map exhibits a poor anomaly–background separation, where the responses of some background regions are intensified. On the contrary, the proposed LELRP-AD method allows all the anomalies to be detected under $c = 0.03$, with fewer background false alarms. The effectiveness of the designed matrix is, therefore, confirmed.

An additional experiment on a simulated data set was also carried out to verify the ability of the proposed method to effectively suppress the sparse background. The simulated data set was based on the ground truth of the Indian Pines data set [42], which contains 17 classes regarded as background classes. Seventeen spectra were extracted from the USGS digital spectral library [43] to represent the background classes. As shown in Fig. 17(b), three simulated anomalies are positioned in the image. Each anomaly has nine pixels, and the total number of anomalous pixels is 27. Another spectrum from the USGS digital spectral library was used to represent the anomalies. The parameters of the experimental methods were set as follows: 1) LSMAD: $r = 2$ and $k = 0.01$; 2) LRASR, $K = 20$, $P = 20$, $\beta = 0.1$, and $\lambda = 0.1$; 3) ADLR: $c = 17$, bw $= 0.2$, and $\lambda = 0.05$; and 4) LELRP-AD: $\rho = 0.2$, $r = 1$, and $c = 0.02$.

The detection results are shown in Fig. 18. The anomalies are not detected in the result of LSMAD since the responses of the anomalous pixels are all zero. Background regions with strong responses exist in the results of LRASR and ADLR. These background regions actually belong to the sparse background since their sizes are all small. On the contrary,

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Fig. 19. Selected local homogenous region.

TABLE III
RANK OF DIFFERENT MATRICES

Tolerance	Rank of the original global data matrix $\mathbf{X}$	Rank of the original local data matrix $\mathbf{G}_k$	Rank of the designed matrix $\mathbf{X}_k$
10^5	3	1	2
10^4	12	6	7
10^3	110	68	68

these sparse background false alarms are effectively suppressed in the result of the proposed method. The AUC scores of LRASR, LSMAD, ADLR, and proposed method are 0.9847, 0.0, 0.9955, and 0.9998, respectively, in which the proposed method achieves the highest AUC score. Therefore, the ability of the proposed method to effectively suppress the sparse background has been verified.

The effectiveness of the "enhanced low-rank prior" idea is now demonstrated. The main idea is that the low-rank property of the designed matrix is enhanced, compared to the original global data matrix. If each spectral vector in the smooth and continuous background in the HSI can be approximately represented as the linear combination of several basis vectors (such as endmembers), then the background can be assumed to hold a low-rank property. If we suppose that the number of background endmembers of the HSI is p_1, then all the background pixel vectors can be represented as a linear combination of these p_1 endmembers, based on the linear mixture model. Thus, the rank of the original global data matrix can be approximately regarded as p_1. Similarly, if we suppose that the rank of the original local data matrix is p_2, then we have $p_2 + 1 < p_1$, since the background classes in the local homogenous region are much less than in the entire image. According to the above inference, $\text{rank}(\mathbf{X}_k) \leq \text{rank}(\mathbf{G}_k) + 1$, and we then have $p_3 \leq p_2 + 1$, where p_3 is the rank of the designed matrix $\mathbf{X}_k$ in the local region. Therefore, we have $p_3 < p_1$. That is to say, the low-rank property of the designed matrix in the local region is enhanced compared with the original global data matrix.

The above conclusion can also be proved from a practical point of view, for which one of the local regions of the AVIRIS-1 data set was extracted, which is highlighted in yellow in Fig. 19, to compare the low-rank property of the original global data matrix $\mathbf{X}$, the original local data matrix $\mathbf{G}_k$, and the designed matrix $\mathbf{X}_k$. The nuclear norm and the rank value under a certain tolerance are adopted to evaluate the low-rank property. The nuclear norm is a convex approximation of the rank of the matrix, and thus, the nuclear norm has a positive correlation with the rank value. The value of the nuclear norms for the original global data matrix $\mathbf{X}$, the original local data matrix $\mathbf{G}_k$, and the designed matrix $\mathbf{X}_k$ is 4.74×10^6, 2.98×10^6, and 3.89×10^6, respectively. The rank values under a certain tolerance are shown in Table III, where the rank of the designed matrix $\mathbf{X}_k$ is always smaller than that of the original global data matrix $\mathbf{X}$. Therefore, the low-rank property of the designed matrix $\mathbf{X}_k$ is enhanced when compared to the original global data matrix $\mathbf{X}$.

Finally, the computational complexity of the proposed method is discussed. The Potts-based segmentation algorithm requires $3p^3q + 3qp$ flops per iteration. The VCA algorithm requires $2r^2N_k$ flops per region. The GoDec algorithm requires $r^2(2N_k + r) + 4B(r + 1)rN_k$ flops per iteration per region. Therefore, the computational complexity of the entire algorithm is $n_1(3p^2q + 3q^2p) + 2mr^2N_k + mn_2\left(r^2(2N_k + r) + 4B(r + 1)rN_k\right)$, where n_1 is the number of iterations for the Potts-based segmentation algorithm, m is the number of segmentation regions, and n_2 is the number of iterations for the GoDec algorithm.

E. Parameter Analysis for the Proposed LELRP-AD Method

Three parameters influence the detection performance of the proposed LELRP-AD method: 1) the scale parameter ρ of the Potts-based image segmentation method; 2) the number of endmembers/rank value r; and 3) the cardinality c. In order to analyze the sensitivity of the parameters for data sets with anomalies of different sizes, the AVIRIS-1 and HYDICE data sets were utilized in the parameter analysis. The AVIRIS-1 data set contains large-size anomalies, and the HYDICE data set contains small-size anomalies. More importantly, these two data sets are widely used in hyperspectral anomaly detection research. The AUC score is used as the quantitative indicator for the analysis. The x-axis represents the range of the parameter being analyzed, and the y-axis represents the AUC score. We analyzed the parameters with the other parameters fixed as appropriate.

To cover the range from oversegmentation to undersegmentation, we set the values of ρ over the range of {0.2, 0.25, 0.4, 0.5, 0.6} and the range of {0.2, 0.3, 0.45, 0.5, 1} for the AVIRIS-1 data set and HYDICE data set, respectively. Accordingly, the numbers of local homogenous regions were {3, 4, 6, 8, 11} and {3, 5, 8, 14, 20}. For the AVIRIS-1 data set, as shown in Fig. 20(a), the AUC score is very low under $\rho = 0.2$, which is because some anomalies are segmented into independent regions when the image is oversegmented, and these anomalies are not detected in the result. However, the AUC scores do not change much as ρ increases from 0.25 to 0.6 even when the image is undersegmented under $\rho = 0.6$. The situation is different for the HYDICE data set, where the AUC score is low when the image is oversegmented ($\rho = \{0.2, 0.3\}$) or undersegmented ($\rho = 1.0$). The reason for this is that the small-size anomalies are sensitive to the sparse background. When the image is undersegmented, the homogeneity of the local regions is destroyed. Therefore, an optimal scale parameter between undersegmentation and

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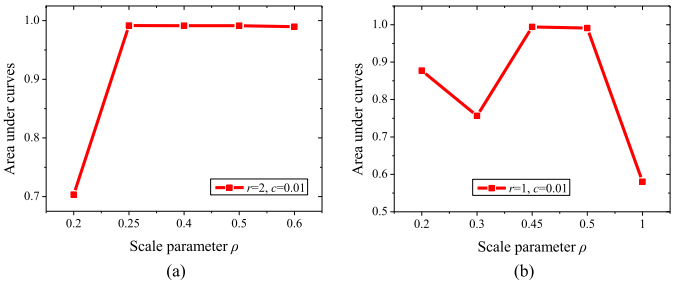


Fig. 20. Analysis of the scale parameter ρ. (a) AVIRIS-1 data set. (b) HYDICE data set.

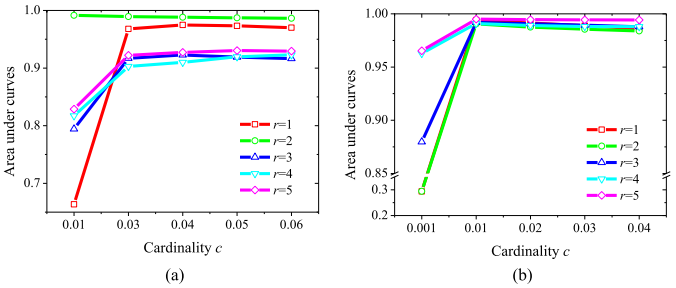


Fig. 21. Analysis of the value of r and cardinality c. (a) AVIRIS-1 data set. (b) HYDICE data set.

oversegmentation is required for accurate anomaly detection with data sets with small-size anomalies. As for the data sets with large-size anomalies, a large range of ρ gives good results. We suggest that the scale parameter ρ is set in the range of [0.1, 0.7] since oversegmentation should be avoided.

For the value of r, as mentioned above, it is highly correlated with the number of dominant background classes in the segmentation region. The minimum value of r is, of course, 1. The number of background classes will not be too large in the segmentation regions, so we set the maximum value of r as 6. Therefore, the values of r were varied over the range of [1, 6]. The cardinality c actually controls the number of nonzero values in the sparse component. A small value of c may lead to some anomalous pixels being undetected, whereas a large value of c makes the sparse component containing more background energy. To cover this range, the values of c were varied over the range of {0.005, 0.01, 0.04, 0.05, 0.06, 0.08, 0.1} and the range of {0.001, 0.01, 0.015, 0.02, 0.03, 0.04}, for the AVIRIS-1 data set and HYDICE data set, respectively. Note that the proportion of anomalous pixels in the AVIRIS-1 data set is larger than that in the HYDICE data set, so the value of c for the AVIRIS-1 data set was set as larger than for the HYDICE data set.

For the value of r and the cardinality c, we performed a joint analysis. As shown in Fig. 21(a) and (b), for the AVIRIS-1 data set, the AUC score exhibits a large variation when the value of r is varied and the value of c is fixed, whereas the AUC score remains stable for the HYDICE data set. There will be an optimal value of r to achieve the highest detection accuracy, but it is not easy to say whether the detection accuracy will increase or decrease as the value of r increases. The optimal values of r are 2 and 5 for the AVIRIS-1 and HYDICE data sets, respectively. This is consistent with the theoretical analysis above, i.e., the number of dominant background classes is small as the segmentation regions hold high homogeneity. Note that the background in some segmentation regions may be relatively complex in some cases, so the empirical range of the value of r is [1, 5]. Furthermore, the analysis was performed with the value of c which is varied and the value of r which is fixed. The AUC scores are quite low under a small value of c because some anomalous pixels cannot be detected. As the value of c increases, there is a critical value that can ensure that all the anomalous pixels are detected. Further increase in the value of c is of no use for the anomaly detection as the sparse component will contain more background energy. It can be seen from the curve that the AUC score does not change much as c increases beyond the critical value where all the anomalous pixels are detected. On the other hand, the value of r affects the value of c to be set. For the AVIRIS-1 data set, $c = 0.01$ results in all the anomalous pixels being detected under $r = 2$. Under other values of r, c needs to be set as a larger value to ensure that all the anomalous pixels are detected. From the above analysis, it can be concluded that the value of r is the dominant factor affecting the detection accuracy, while the cardinality c has little influence. Therefore, the empirical range for the cardinality c is [0.01, 0.03], on the condition that the value of r is set appropriately.

V. Conclusion

In this article, inspired by the observation that, in local homogeneous regions, background signals hold an enhanced low-rank property while anomalies exhibit spatial sparsity, and a novel hyperspectral anomaly detection method has been proposed based on an LELRP-AD. The proposed LELRP-AD method achieves anomaly–background separation while avoiding the need for a complicated dictionary construction process. The image is first decomposed into a set of local homogenous regions, in order to enhance the low-rank property of the background signals, where a Potts-based image segmentation method is adopted with postprocessing. The original local data matrix is then augmented with the extracted background endmembers for the low-rank and sparse decomposition, ensuring anomaly–background separation. Finally, the anomaly detection map is obtained by combining the sparse components of all the local regions. Four real-data experiments confirmed the superiority of the proposed LELRP-AD method in both the quantitative and qualitative performance, compared with the classic and state-of-the-art low-rank-based hyperspectral anomaly detection methods.

The experiments showed that the scale parameter ρ hardly affects the detection performance for a data set with large-size anomalies, whereas the results for a data set with small-size anomalies are sensitive to this parameter. Furthermore, it was found that the proposed method is not greatly dependent on the value of r and the cardinality c, and good detection results can be obtained over a large range of r and c, especially for the cardinality c. We suggest that the scale parameter ρ is set in the range of [0.1, 0.7], to avoid oversegmentation. As for the number of endmembers/rank value r and cardinality c, the range

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of c that we suggest is [0.01, 0.03]. Note that the local regions hold significant homogeneity so that we suggest that the value of r is set as a small value, and the empirical range is [1, 5]. If a higher detection accuracy is required, these parameters can be adjusted, according to the given range. In the future work, more UAV HSIs [44], denoising process [45], and other computational intelligence methods [46], such deep learning, will be considered for the target detection of hyperspectral remote sensing imagery.

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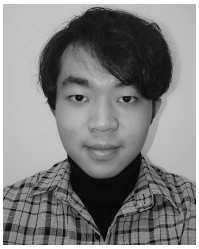
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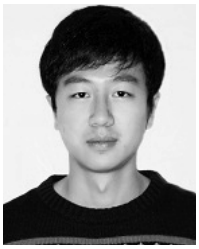
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